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Connecting LibreOffice Calc with Base and MariaDB/MySQL

This article demonstrates how using LibreOffice Calc with Base and a Structured Query Language (SQL) database like MariaDB or MySQL can help anyone who wishes to keep track of mutual fund investments. The convenience afforded is amazing!

Let's assume that you have a list of mutual funds that you can easily manage in a spreadsheet. You will need to check the NAV (net asset value) of the funds occasionally to keep track of your assets. You can find out the NAV of all the mutual funds (MFs) you have invested in at one convenient spot—the website of the Association of Mutual Funds, [amfiindia.com](http://www.amfiindia.com).

As time goes by, your list of funds keeps increasing and at some point you will want to automate the process of keeping track of them. The first step is to download the NAV for all the funds in a single text file; you may use *wget* to do so:

```
$ wget http://www.amfiindia.com/spages/NAVA11.txt
```

The downloaded file is a text file with fields separated by ','. It can easily be loaded into a spreadsheet and examined.

Creating a usable format

Each mutual fund can be identified by an International Securities Identification Number (ISIN). Hence, ISIN (the second column) can be the key to organise the data. The minimum additional data you need is the NAV for the day (the fifth column) for each security.

LibreOffice has a Base module, which I had never used and which works well with any SQL database using ODBC/JDBC drivers. If you are using Linux, you may already have installed MariaDB or MySQL. Python will easily work with MySQL as well.

So, you create a MySQL database NAVDB and a table called *navtable*:

```
CREATE TABLE navtable (isin char(12) primary key, nav
real);
```

The following code is a minimal Python program to load the data into a MySQL table while skipping all the extraneous rows and rows for which no NAV is available.

```
import MySQLdb
SQL_INSERT="INSERT INTO navtable(isin,nav) VALUES('%s','%s)"
db = MySQLdb.connect("localhost","anil","secretpw","navdb")
c = db.cursor()
try:
    for line in open('navall.txt').readlines():
        cols=line.split(';')
        if len(cols) > 4 and len(cols[1])==12 and not
cols[4][0].isalpha():
            c.execute( SQL_INSERT%(cols[1],cols[4]) )
        db.commit()
    db.close()
except:
    db.rollback()
```

Linking the MySQL database and LibreOffice Base

In addition to LibreOffice Base, you will need to install the MySQL JDBC driver. On Fedora, the package you need is *mysql-connector-java*.

LibreOffice needs to be able to find this driver. For Fedora, add the archive */usr/share/java/mysql-connector-java.jar* in the class path of *Java Options*. You can find this option in *Tools/Options/LibreOffice/Advanced*.

The next step is to enable this as a data source in LibreOffice, as follows.

1. Create a new database.
2. Connect to an existing database using MySQL JDBC connection.
3. The database is *navdb* on localhost.
4. Enter the credentials to access the database.
5. You will be prompted to register the database with LibreOffice.
6. You will be prompted for the password.

You may create additional tables, views, and SQL queries on the database, which may then be accessed through Calc.

Access the data through Calc

Your personal table of your funds will have at least the